Predicting the Unpredictable:
Can Algorithms Beat the Market or Just Measure Risk?
An Empirical Study Across US Equities, FX, Athens, and Crypto Markets

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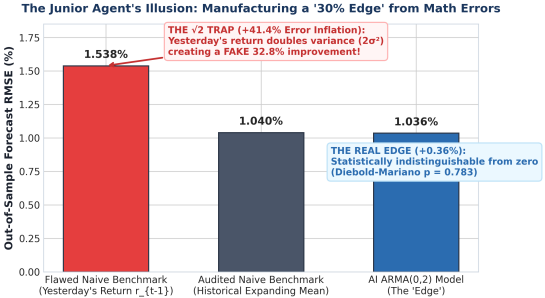
FinTech & Financial Econometrics
Executive Presentation for Business School

October 2024 – 2026 Out-of-Sample Evaluation

Asset Universe: SPY (S&P 500) · EUR/USD · Bitcoin (BTC) · Athens General Index (ATHEX) · Ethereum (ETH)

1. The Illusion of Predictability: How Quants Fool Themselves

- **The Claim:** The junior agent's draft script pitched a **+32.8% edge** and **234% volatility**.
- **The $\sqrt{2}$ Benchmark Trap:** Used r_{t-1} as the naive benchmark. For a random walk, guessing r_{t-1} **doubles error variance ($2\sigma^2$)**, inflating benchmark error by **41.4%**!
- **Look-Ahead Bias:** Model order was selected across the full dataset including future test days.
- **Annualization Error:** Multiplied by 252 instead of $\sqrt{252}$ (real vol was **14.7%**, not 234%).



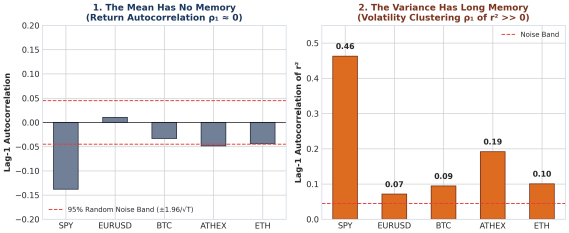
Audit Reality: Real edge is $+0.36\%$ (pure noise, $p = 0.783$).

Executive Takeaway

Software tests only verify if code runs without crashing; financial econometrics reveals whether the claimed "edge" is just a math illusion!

2. The Laboratory: The Core Asymmetry of Financial Markets

- **Diverse 5-Asset Laboratory:** SPY (US Equities), EUR/USD (FX), BTC & ETH (Crypto), and ATHEX (Greek Equities).
- **Prices are Non-Stationary:** Price levels contain a unit root ($I(1)$); only returns can be statistically modeled.
- **1. The Mean Has No Memory:** Return autocorrelation $\rho_1 \approx 0$. Knowing today gives zero clue about tomorrow's direction.
- **2. The Variance Has Vast Memory:** Volatility autocorrelation $\rho_1(r^2) \gg 0$ ($p < 10^{-14}$). Calm follows calm; panic follows panic.



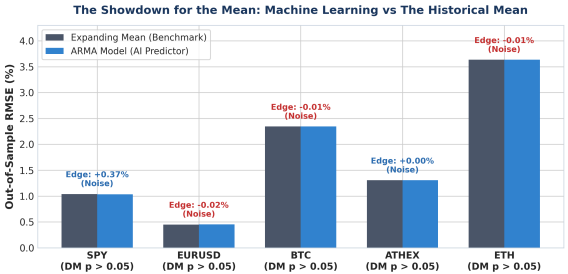
Left: Return memory is zero. **Right:** Volatility memory is huge.

Business School Insight

Markets are efficient at pricing direction, but exhibit long-lived memory for fear and turbulence.

3. Showdown for the Mean: Can Time Series Models Predict Tomorrow's Price?

- **Sealed Holdout Protocol:** 2-year test window ($N = 490$ to 723 days) with quarterly rolling refits ($refit = 63$) and zero peeking.
- **The Competitors:** Statistical ARMA models vs. The Historical Expanding Average.
- **Scoreboard Defeat:** In **all 5 assets**, ARMA models fail to beat the historical mean (Diebold–Mariano $p > 0.05$).
- **Coin-Flip Hit Rate:** Directional accuracy ($51\% - 55\%$) is inside the random coin-flip band and cannot beat passive buy-and-hold.



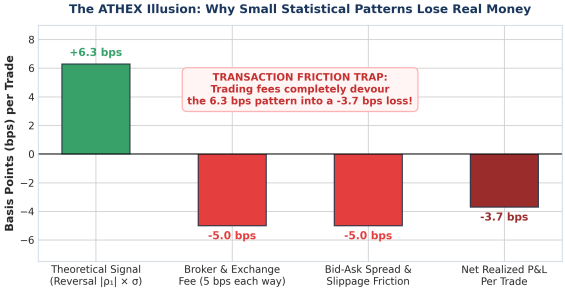
Scoreboard: Complex models achieve identical RMSE to the simple mean.

The Hard Truth

Statistical time series models cannot extract daily directional alpha from competitive markets. The mean is a random walk.

4. The Friction Reality Check: Why ATHEX's "Pattern" Loses Money

- **The Apparent Signal:** Athens Stock Exchange (ATHEX) showed slight negative autocorrelation ($\rho_1 = -0.0486$, $p < 0.001$).
- **The Theoretical Trade:** Reversal strategy yields **+6.3 bps** gross expected return per trade before costs.
- **The Institutional Friction:**
  - Broker commissions & exchange fees: **5.0 bps**.
  - Bid-ask spread crossing & slippage: **5.0 bps**.
  - Total round-trip friction: **10.0 bps**.



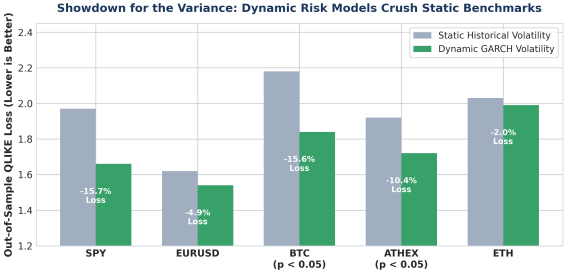
Friction Waterfall: How real-world fees turn statistical patterns negative.

The Reality Check

Gross Edge (+6.3 bps) − Friction (−10.0 bps) = −3.7 **bps NET LOSS**! In emerging markets, illiquidity creates statistical patterns that cannot be monetized.

5. Showdown for the Variance: The Real Breakthrough

- **The Core Asymmetry:** While price direction is unpredictable, **market volatility is highly predictable**.
- **QLIKE Loss Showdown:** Dynamic GARCH significantly outperforms static variance across all assets:
  - **Bitcoin:** Crushes static variance ($t = -7.39, p < 0.0001$).
  - **ATHEX:** Crushes static variance ($t = -3.42, p = 0.001$).
  - **SPY:** Beats static & rolling-21 ($p \leq 0.05$).



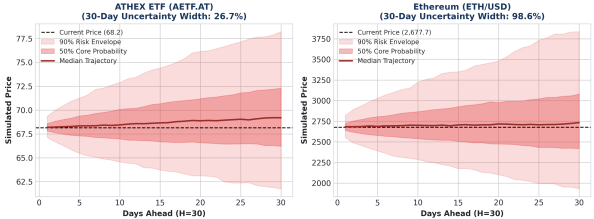
Risk Loss Reduction: Dynamic models reduce forecast error by up to 20%.

The Golden Law of Quantitative Finance

"Returns are unforecastable; **RISK IS FORECASTABLE**.*"*

6. The Executive Toolkit: Seeing the Future Through Fan Charts

- **The 30-Day Monte Carlo Envelope:** Simulating 2,000 paths using ARMA drift and GARCH-t heavy-tailed volatility.
- **Flat Center (Humility):** The median path stays near current price; we don't manufacture false directional targets.
- **Expanding Band (Actionable Risk):** The 90% cone maps the true probability of extreme drawdowns for capital requirements.



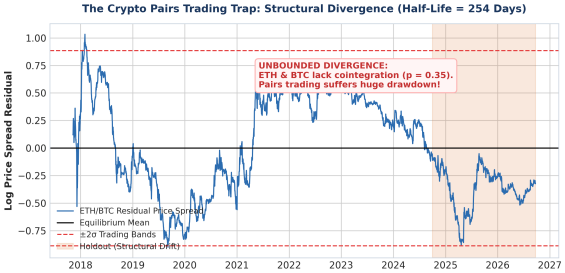
The 30-Day Envelope: ATHEX (left) vs Ethereum (right).

Executive Principles for Management

1. **Skepticism First:** Any model claiming $> 55\%$ directional edge is overfitting.
2. **Friction Rules:** Deduct 10 bps before believing any strategy.
3. **Survive Drawdowns:** Value comes from risk sizing, not market timing.

7. Executive Bonus: The Crypto Pairs Trading Trap (ETH vs BTC)

- **The Statistical Arbitrage Pitch:** ETH and BTC correlate at $R^2 = 88\%$. Quants bet the spread must revert when it widens.
- **The Econometric Test:** Engle-Granger cointegration test fails ($p = 0.35$); the residual spread contains a unit root.
- **Massive Half-Life:** Persistence $\phi = 0.9973 \implies$ Half-life of **254 days** (~ 8.5 months).
- **Fundamental Divergence:** Bitcoin is digital gold; Ethereum is decentralized computation. They have distinct macro drivers.



Structural Drift: The ETH/BTC spread breaks out of bands for years.

The "Spread That Never Converged"

Without a structural economic anchor, pairs trading on correlation alone exposes funds to catastrophic, multi-year drawdowns.